Ph.D. Dissertation Proposal

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Pertinent Information

Proposed Title: Acoustic Detection of Unmanned Aerial Vehicles and Insects in Stored Products Leveraging Relational Databases to support Multi-modal and Multi-sensor Data

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1 Introduction

Acoustic sensing has been a reliable and relatively low-cost alternative to even the most advanced sensing technologies to provide a method of detecting, localizing, tracking, and classifying subjects including cracks in wind turbine blades, marine mammals and scuba divers underwater, and low-flying aircrafts [1–3].

Large amount of acoustic data recordings are required not only to study the characteristics of the acoustic signals generated by the studied subjects, develop algorithms, validate sensor performance, but also to support the increased advancements and usage of machine learning and artificial intelligence. While the research community has access to vast repositories of text, image, and video datasets that have driven significant advances in learned language modeling (LLM) and computer vision, acoustic datasets remain comparatively scarce [4–6]. This might be attributed to visual data modalities such as images and video are easily annotated and inherently interpretable due to their direct representation of observable phenomena. Acoustic data presents a more abstract signal that necessitates corroborative information including subject characteristics, environmental parameters, and sensor specifications to enable meaningful analysis and classification.

This work focuses on the demonstration of the possibilities of acoustic sensing to detect unmanned aerial vehicles and insects within stored products by using simple signal processing techniques with promising performance and avoiding the need for machine learning and artificial intelligence. On the surface, these targets seem superficially different from their size and function to their acoustic signatures. Conversely, their audible existance with unique sonic characteristics make them opportune canditates for acoustic tracking. Furthermore, they both pose a threat to national security, drones as they become more prelevant for espionage, smuggling, and as weapons or insects as they affect food safety, agriculture, the environment, or enter delicate ecologies as invasive species.

In the first section, the acoustic detection of unmanned aerial vehicles (UAVs) is discussed. The section presents an updated prototype to the previously developed Drone Acoustic Detection System (DADS). The Drone Acoustic Recording Array (DARA) is a seven-microphone system compared to the four-microphone system of the DADS. Both sensors were deployed over the course of two field tests where numerous drones of different sizes and amount of propellors were flown. After comparing the spectrograms of the UAVs and finding their sound pressure levels (SPL) at 1 m, an algorithm was developed to predict the detection range of each sensor at different noise levels and environtal conditions using wideband processing and the sonar equation. The actual detection algorithm utilizes the Steered-Response Phase Transform (SRP-PHAT) to localize UAVs [7].

The following section examines the acoustic detection of insects in stored products. The Acoustic Stored Product Insect Detection System (A-SPIDS) was developed to address the need for an efficient alternative to manual visual inspection of materials such as grain, rice, and legumes for insects at United States Ports of Entry. The sensor was used to collect an extensive amount of recordings using live insect specimen at various life stages in materials of different kernel size and hardness with varying levels of natural and simulated background noise. This data was used to study the characteristics of the impulses generated by the insects as they manuevered around the material and derive metrics based on the pulse amplitude and spectra. A method of normalization was also introduced that could extend to other insect sensors. Utilizing these normalization and filtering techniques, a detection algorithm was developed to effectively detect the presence of a single insect within 1 L of material with up to 90 dBA external noise [8].

Underlying both of these studies is a database package that leverages relational database structures to organize and store the large amounts of acoustic data along with their respective corroborative information. The Sound Organization and Network Integration for Collection/Collaboration Database (SONICDB) was designed to be flexible and extendable, allowing for its adoption to service projects of varying needs. The Stored Product Insect Database (SPIDB) is a simple extension of the SONICDB package, only adding models to support logging of the insect subjects in their paired materials. In the Drone Acoustic Dataset, the SONICDB package supports data from five field tests, incorporating acoustic data from 14 different sensors recording a total of eight different drone models. Additionally, the dataset also features flight logs, GPS truth data, ADS-B, and weather data.

2 Acoustic Detection of Unmanned Aerial Vehicles

2.1 Literature Review

Unmanned Aerial Vehicles (UAVs), also known as drones, have become prelevant in consumer, commercial, industrial, and defense industries. Although their usage has entertained families, assisted farmers to monitor and service large amounts of agricultural areas, and enabled first responders, drones have also been used in nefarious purposes such as surveillance, smuggling, and as weapons [9]. The weaponization of UAVs has been highlighted by the conflict in the Russian-Ukraine war demonstrating the efficacy of low-cost drones to monitor the battlefield, deliver explosives, jam communications, and even conduct kamikaze-style attacks [10]. To combat this threat, Counter-Unmanned Aerial Systems (C-UAS) have been developed to detect, localize, track, classify, and mitigate using sensing technologies including radar, passive radio-frequency, electro-optical (EO) and infrared (IR) cameras, and acoustics. Although acoustic sensors provide a relatively lower detection distance than their radar and RF counterparts, they are comparitevely lower cost, passive, and are capable of detecting, localizing, tracking, and classifying UAVs [11].

Diaconu and Petrița demonstrated that a standard off the shelf sound card with a four microphones array successfully estimated the direction of arrival within 5° [12].

Several drone focused datasets are publically available for research. Although the Center for a New American Security (CNAS) developed the Drone Database as a part of the Proliferated Drones project with the National Security Program, this dataset has not been updated regularly [13]. There are several datasets available for imagery and radio-frequency based drone detection [14–17].

Acoustic data requires a rich set of metadata to provide context beyond the identification of the platform within the recording. This could range from information about the recording system such as the microphone type and placement, microphone array geometry, frequency response, and sensitivity ranges across recordings. Environmental information such as weather conditions and temperature or background noise that could include the presence of other aircrafts, vehicles, and people. Details about the flights such as the distance between the recording device and platform, the flight pattern and manuevers, the altitude, and speed. This minutiae enables the development and testing of algorithms and models to be robust to the various conditions encountered during deployment. Even when the data is provided, it is often decoupled and requires significant effort to match the audio.

Acoustic drone datasets often provide data in the form of raw audio recordings which enable researchers to develop and test their own algorithms and models. The DronePrint dataset features over 100 minutes of audio recordings, mainly from videos available from YouTube, in a variety of noise environments and was used to generate a machine learning pipeline for drone detection and classification [18]. This dataset does not feature any other metadata [19]. Conversely, some datasets provide processed data, such as the Comprehensive Dataset for Acoustic Drone Detection which shares the data for 28 unique drone models only as processed mel-spectrogram images. The dataset authors do not provide any information beyond the manufacturer and model of the drone as well as the signal processing techniques used to generate the spectrograms [20, 21].

Some datasets feature ground based platforms recording the UAVs in flight. The DroneAudioDataset features recordings of the Parrot Bebop, Parrot Mambo, 3DR Solo, DJI Phantom 4, and AR Drone made with the iPhone's integrated microphone along supplemental information including bearing to target, range to target, target altitude, and ambient temperature. This metadata is stored in the filename under different indices requiring custom code to parse [22–24]. **yiSoundbasedDroneFault2023** released 7.6 GB of audio recordings from a two-microphone system capturing three different hobbyist sUAS platforms mounted stationary in an anchoic chamber and performing flight manuevers with different physical fault conditions on the propeller and motors. The authors mixed the audio with noises recorded in five different areas on a university campus.

Other datasets feature recording systems mounted directly on the flying platforms. The DRone EGonoise and localizatiON (DREGON) dataset was designed to further research on UAV-embedded sound source localization. The dataset features audio recordings, ground truth relative source position, motor speed logs,

and inertial measurement unit (IMU) logs of a MikroKopter quadrotor drone flying an anechoic chamber with an eight microphone array is mounted directly on the frame. Although the supplimentary metadata is extensive, the data is decoupled into seperate files and would need to be fused manually [25, 26]. Dümbgen et al. published acoustic alongside their research on audio-based navigation using microphones mounted on a quadcopter. The dataset also features platform mechatronic information, such as rotor speed and pitch, it is also unlinked and would require effort to match the audio. Although Wang et al. boasts of a extensive audio dataset featuring ten different UAVs that was used for several publications, the dataset is not publically available at the publication time of this paper [20, 29, 30].

The Sensor Technology and Applied Research (STAR) Center laboratory at Stevens Institute of Technology applied the tools and technologies utilized in low-aircraft detection and iterated the designs to produce the Stevens Circular Acoustic Array (SCAA), the Drone Acoustic Detection System (DADS), and the Drone Acoustic Recording Array (DARA) [7, 37, 41].

2.2 Motivation

As drone usage becomes more pervasive, the need for effective detection and tracking systems has grown. Although the relatively low detection distance and susceptibility to environmental noise are a challenge for acoustic sensors, they still offer a promising solution due to their low cost, passive nature, and weather resistance especially when deployed in a networked and fusioned system with other modalities. By publishing this research as well as the data collected, it is hoped that it will encourage further development and refinement of algorithms, platforms, and usage within machine learning models. In the 2025 One Big Beautiful Bill, the United States allocated $2 billion USD for research, development, and procurement of counter-Unmanned Aerial Systems [42].

2.3 Methodology

3 Acoustic Detection of Insects in Stored Products

3.1 Literature Review

Stored products refer to raw and processed non-perishable commodities, including grains, legumes, nuts, spices, and manufactured food. Fundamental for food security and nutrition, they are also economically valuable and represent a large portion of global trade. These products are susceptible to infestation by various pests, resulting in quantitative and qualitative losses as well as pose a risk to human health and safety. Efficaciously inspecting products for pests throughout the global supply train is not only crucial for the integrity of the product and mitigate greater infestation but also secure regional agricultural and environmental ecologies from the introduction of invasive species.

3.1.1 Stored Product and Insect Pests

Grains include cereals, pulses, and oilseeds. Cereals are grasses cultivated for its edible grain such as wheat, rice, corn, rye, millet, and barley [43]. Pulses are the edible seeds of the *Leguminoseae* family and include beans, lentils, peas, and chickpeas. Peanuts, soybeans, and alfafa are legumes but are considered to be oilseeds, such as sunflower seeds, rapeseeds, and sesame seeds [44].

Insects attack stored products in a variety of ways and are characterized by their feeding behavior. The rice weevil, *Sitophilus oryzae*, the granary weevil, *Sitophilus granarius*, and the lesser grain borer, *Rhyzopertha dominica*, are an example of internal feeders, which bore into the kernels of grains and seeds as larvae and feed on the endosperm, visually undetectable, until they emerge as adults leaving burrowing holes and powder residue. External feeders, such as the drugstore beetle, *Stegobium paniceum*, the cigarette beetle, *Lasioderma serricorne*, and the Khapra beetle, *Trogoderma granarium*, chew through the outer seed coat of the grain exterior and then devour the inside. The red flour beetle, *Tribolium castaneum*, and the sawtoothed grain beetle, *Oryzaephilus surinamensis* are examples of scavenger feeders, which feed on broken down and processed material such as broken kernels and flour. Secondary pests, like the mealworm, *Tenebrio molitor*, eat deteriorating, damp, or moldy material [45].

Stored product pests (SPPs) can infest crops at any point from the field to storage. Infestations lead to both quantitative and qualitative losses. Direct consumption of grain reduces its weight and volume, while contamination from excrement, webbing, and carcasses degrades its quality. By consuming grain embryos, insects lower the protein content and alter the chemical composition of the kernels. This damage promotes infections, attracts other pathogens, and reduces seed viability. Furthermore, SPPs can damage storage containers and equipment [46]. The rate of infestation is accelerated by high temperatures, and the moisture and heat generated by the insects create an even more favorable environment for their proliferation [47].

An unmitigated infestation can lead to significant ecological, agricultural, and economic impact. Estimates by the Food and Agriculture Organization (FAO) of the United Nations suggest that 20 % of the global food production is lost to herbivorous insects [48]. Other sources state that post-harvest insects are responsible for 10 % to 20 % of storage losses [49, 50] and cost up to $470 billion USD per year in 2001 [51]. Introduction of alien insects into ecological enviornments can further exacerbate negative affects if the species has no natural predators. Invasive species cost the United States $2 trillion USD from 1960 to 2020, costing $1 billion USD a year from 2010 to 2020 in damages, losses, and prevention [52].

3.1.2 Methods to Inspecting Stored Product for Insect Infestation

According to the International Organization of Standards (ISO) 6639 standard, a container is determined to be infested if a single insect is found within 1 kg of material [53]. The United States Department of Agriculture (USDA) also guides the Customs and Border Protection (CBP) Agriculture Specialists (CBPAS) to deem a container infested if one or two insects are found within 1 kg of material sample.

The main method employed at United States Ports of Entry to inspect stored product for insects is through manual inspection, where the chosen commodity is emptied on an inspection table and personnel, often with

tools including sieves, tweezers, magnifying glasses, and intense lights sift through the material looking for live and dead insects at different life stages, fragments, or signs of insect infestation such as webbing, casings, frass, or burrowing holes in the kernels. This process is time and physically consuming, requiring experienced personnel or with a background in entomology to identify the species of the insect in order to determine the appropriate course of action. If the insect is identified as a quarantine pest, the material is either treated, destroyed, or rejected and a Emergency Alert Notification (EAN) is issued to the importer [54].

Various tools and devices have been developed, often leveraging sensing technologies, in order to expedite areas within the inspection including automating the examination process, detecting insect infestation, or identifying the insect species.

Physical methods use probe traps and counters, such as the Berlese-Tullgren trap, the TNAU Insect Detection System, and the Electronic Grain Probe Insect Counter (EGPIC), to enumerate the captured insects and determine the level of infestation using extrapolation. Some probe traps can be placed at different grain depths for extensive amounts of time and periodically removed to evaluate the number, species, and density of the infestation. Although low-cost, passive, and effective at long-term storage facilities, this method is restricted to easily accessible locations within the container and the capture rate is influenced by the insect species, grain type, as well as temperature and humidity. Although visual lures and pheromones assist in attracting insects to the traps, these physical methods require a long time duration and cannot detect insects buried inside kernels [55, 56]. Other physical methods include rolling machines that automate the sieving of products, systems that monitor for changes in electrical conductance between whole, damaged, and infested kernels, and the flotation method which separates possible insects from product by density, but these tend to damage the material in the process [57–59].

Chemical sensors monitor the O_2 and CO_2 levels in the stored product to detect the respiration of the insects or by measuring for uric acid, the main element of insect exreta [60–62]. Even though it was not developed for insect detection, the electric nose (E-nose), which consists of gas sesnors with varying sensitivies and pattern recognition software, was able to detect the volatile compounds and react to the changing electrical properties to identify an infestation of the lesser grain borer, *Rhyzopertha dominica* in wheat [61, 62].

Advanced imaging technologies offer non-destructive solutions for pest detection. By using an X-ray source and imaging system, inspectors can visualize insects concealed within stored grains, fruits, and vegetables, as well as find damaged products [63, 64]. Alternatively, near-infrared (NIR) spectroscopy leverages the unique light absorption properties of materials to detect insects and classify infested kernels, though its need for constant recalibration is a drawback [**keszthelyiNonDestructiveImagingSpectroscopic2020a**, 55, 65]. Thermal imaging detects the infrared energy emitted by objects, creating heat maps that expose infestations through temperature anomalies [66]. To boost precision, these systems are often paired with deep learning algorithms [46].

Mankin demonstrated that a 24 GHz microwave Doppler radar, originally used for termite detection, can also find stored product pests. The insect movement is detected by radiating a microwave signal and observing the change in the frequency of the received reflection, which increases as the insects move slightly closer or decreases as they move further away from the sensor. The system was able to detect the cigarette beetle, *Lasioderma serricorne*, the sawtoothed grain beetle, *Oryzaephilus surinamensis*, the black carpet beetle, *Attagenus unicolor*, and the red flour beetle, *Tribolium castaneum* within a corn meal and flour mixture[67].

3.1.3 Acoustic Methods and Algorithms for Detecting Insects in Stored Products

Insects generate sounds within the material during locomotion through friction between the body and the material as well as through the movement of the wings, legs, and other body parts. Sounds are also generated by mastication, communication, and procreation. The production and magnitude of these sounds are strongly affected by the insect size and lifestage, feeding habits, speed, activity, the density of the infestation, and the general behavior of the subject which can vary with temperature and external disturbance. The propogation of the sound is dependent on the material properties including density, kernel size and hardness, relative humidity, and the amount of air in the material [68]. The distance between the insect and the sensor as well as variation in the recording equipment and environmental conditions can also introduce differences in the

results from different studies and publications. Often the sensors, signal features, processing, and algorithms are common with the platforms, analysis, and detection methods for insect infestations in other materials such as plants, soil, and wood [39].

The insect activity manifests in acoustic recordings as impulses, high magnitude signals across a wide frequency range and short duration which can be characterized by their intensity, duration, and spectral features. Microphones, piezoelectric sensors, laser vibrometers, micromechanical sensors (MEMS), accelerometers, ultrasonic transducers, and seismic sensor have all been effective at capturing these features to be used by detection algorithms to determine the presence of an insect, monitor for hidden infestations without destroying the product, classify specific species and their respective life stage, quantify the magnitude of the infestation, as well as mitigate the influence of external noise.

The spectral domain refers to the frequency content of a sound. Generally, insect noise occurs in the frequency band between 80 Hz to 8000 Hz [68] and can vary between the species and lifestages [69, 70]. The khapra beetle, *Trogoderma granarium*, in wheat grains was most dominant in frequencies 1000 kHz to 8000 kHz. The adult and larva occupied different bands, 3500 Hz to 4500 Hz and 1000 Hz to 1700 Hz, respectively. Female adults resonated mainly in the 1000 Hz to 3000 Hz and 4000 Hz to 5000 Hz frequency bands [69]. The peak energies for an adult and larva of the rice weevil, *Sitophilus oryzae*, in wheat grains were within 1800 Hz to 3000 Hz and 1300 Hz to 2000 Hz, respectively [70]. This variety in frequency content enabled Schwab et al. to distinguish between the larval and adult stages of the *Sitophilus granarius*, *Tribolium confusum*, and the *Rhyzopertha dominica* through spectral features alone.

The temporal domain refers to the time-varying nature of the signal including amplitude and duration. A single impulse might be 3 ms to 30 ms. Bursts, which are a series of pulses that can be enumerated by their number and rate within a time period, can range from 50 to 2000 impulses for a duration from 3 ms to 30 ms with up to 250 ms intervals of seperation. Adult lesser grain borers, *Rhyzopertha dominica*, and red flour beetles, *Tribolium castaneum*, produced 37 and 80 times more sound than their larvae [68]. An increase in impulses could be indicative of a higher population infesting the material [72]. Some algorithms count the number of bursts and impulses to determine an infestation [68, 73–75].

Signal processing such as filters and transformations can be used to isolate the frequency range of interest and expose the relevant signals while removing superfluous noise from analysis that can vary between sensors and environments. The waveform envelope allows for threshold crossing algorithms to determine an insect precense if the amplitude exceeds a statitically or empirically determined cutoff [75]. Mankin et al. observed that the sounds of *Sitophilus oryzae* movement and feeding were 23 dB. Balingbing et al. continously recorded the *Rhyzopertha dominica*, the *Sitophilus oryzae*, and the *Tribolium castaneum* in rice over a 24 h time frame using an Adafruit 12S MEMS SPH0645 microphone and calculated the root-mean-square (RMS) of 3 min windows at each hour of a signal filtered between 93.75 Hz to 2500 Hz. Eliopoulos et al. utilized a 14 cm long piezoelectric sensor inserted with a probe into hard wheat grain. The system was able accurately detected the presence of an insect with a probability range from 72 % to 100 %, a 100 % false positive rate. Their algorithm was based on extracting the Hilbert envelope and indicating a detection when it meant a certain threshold.

The regularity and duration of the monitoring also has an effect on the performance of the sensor and its algorithm. Vick et al. noted that the *Rhyzopertha dominica*, *Sitotroga cerealella*, and *Sitophilus oryzae* produced sounds 61 %, 71 %, and 90 % of the time in grain during sampling durations of 180 s. Short analysis durations can increase high rates of false negatives especially if the insects are not active during the sampling period. Longer sampling periods, such as 10 min to 20 min allow for a more accurate detection as well as enable identification of background noise [68]. Deployment of multiple sensors simultaenously could increase detection performance, determine locations of insects, level of infestation, and to offset the frequency and duration of monitoring [68]. Hagstrum et al. concluded that the frequency of monitoring a single sensor had an equivalent performance as monitoring multiple sensors at a lower frequency.

Limiting the influence of external noise and vibration from the surrounding environment, such as factory machinery, vehicles, and other sources, is critical to the success of these algorithms. External noise can mask the low-intensity sounds produced by insect feeding and movement, which typically range from 20 dB to 80 dB SPL (sound pressure level) at frequencies below 10 kHz. Often this background noise can be reduced

physically through shielding, sound-proofing, and vibration isolation. [76] was able to detect internally feeding larvae in grain samples at an inspection facility with high noise backgrounds by shielding the acoustic detectors using a multi-layered enclosure that attenuated sound by 70 dB to 85 dB between 1 kHz to 10 kHz [76]. Other methods could be through box in a box design, anechoic chambers, employing Faraday cages, and shock mounts. Alternatively, reference sensors could be used to measure and identify peripheral noise and vibration [73].

The spectral and temporal differences between the insects and background noise can also be used to distinguish between the two. Mankin et al. used filtering to reduce signals below 200 Hz. In another algorithm, the bursts characteristic to distinguish the insect sounds from the continuous or isolated impulses commonly found in background noise [80, 81].

Machine learning algorithms can analyze acoustic signals and identify patterns associated with insect activity as well as distinguish external noise however they depend on large amounts of training data. The ML models can be trained on the raw audio signals or on extracted features and the data can be provided labelled or unlabelled [82]. These algorithms include various nueral network classifiers such as convolution neural networks (CNN), probabilistic neural networks (PNN), perceptual learning prediction (PLP), decision trees and forests, hidden Markov models (HMM), support vector machines (SVM), and bayesian classifiers [75]. Deep learning combines multiple layers of neural networks to learn the features and patterns of the data. Since machine learning algorithms are data-driven, the performance of the model might not generalize to new species or materials, if the training data is not diverse enough.

Phung et al. used the BugBytes dataset to compare training a linear discriminant, fine kNN, linear SVN, and Bagged Tree model using low-level features such as the zero-crossing rate, short-time average energy, bandwidth, and cut-off frequency as well as the mel-frequency ceptstrum coefficients (MFCC). They concluded that the Bagged Tree model provided the best accuracy for species classification and insect classification with an accuracy of 97.1 % and 92.3 %, respectively while using both low-level and MFCC features. Montemayor et al. used the mel-frequency cepstrum coefficients from the audio signals and their log-scaled representations as Time-Frequency Patches (TF-Patches) as inputs to a Convolutional Neural Network (CNN). Faiß and Stowell observed that training a LEAF model with raw waveform data from the diverse InsectSet32, InsectSet47, and InsectSet66 resulted in better performance than a CNN model with spectrograms, a graphical visualization of the frequency magnitude over time.

The Acoustic Location Fixing Insect Detector (ALFID) system was designed to be vertically mounted with grains gravity loaded and unloaded. The system utilizes an array of sixteen linearly oriented piezoelectric sensors ranged to identify the number of feeding larvae and their location within the sample. After amplifying the signals from the sensors by 80 dB, a bandpass filter from 1000 Hz to 10 Hz is applied. The arrival time of the signal is identified using amplitude threhold detection and the location is determined by a detection order algorithm.The system was able to detect the presence of up to three rice weevil, *S. oryzae*, larvae in 1 kg of wheat with a detection probability of 64 % and a false positive rate of 8 % [86, 87].

The Beetle Sound Tube was developed to support semi-permanent monitoring of silos and flat stores of grain to detect low levels of infestation early and provide remote reports. The system comprises of perforated metal tubes that are installed into the grain mass. Each tube features an insect trap that can be inspected for species classification, an acoustic sensor, and a climate sensor to measure changes in temperature and humidity. The signal processing on the acoustic signals utilize $\frac{1}{3}$-octave analysis, calculating the energy level for 16 $\frac{1}{3}$-octave bands from 250 Hz to 8000 Hz. The algorithm determined an insect was present if the energy level exceeded a threshold of 25 dB within 250 Hz to 800 Hz, 20 dB within 800 Hz to 2500 Hz, or 15 dB within 2500 Hz to 8000 Hz, above the background noise. The sensor and algorithm was able to detect the sawtoothed grain beetle, *Oryzaephilus surinamensis*, within barley and was able to determine the level of infestation using cross-correlation [47].

The Acoustic Emission Consulting AED 2010L was able to detect one to two insects per kg of wheat grain with 72 % to 100 % accuracy. The portable, battery-operated system uses a piezoelectric sensor mounted on a metal probe that is inserted into a grain sample. The system was able to predict the population densities of the tested live specimen including the rice weevil, *Sitophilus oryzae*, the lesser grain borer, *Rhyzopertha dominica*, the confused flour beetle *Tribolium confusum*, the sawtoothed grain beetle, *Oryzaephilus surina-*

mensis, the rusty grain beetle, *Cryptolestes ferrugineus*, the khapra beetle, *Trogoderma granarium*, and the cigarette beetle, *Lasioderma serricorne*, through chi-squared tests and machine learning classifiers [78, 88].

The Postharvest Insect Detection System (PDS) uses electret microphones centered on a plexiglass base onto which grain sample is placed. The system was tested with the rice weevil, *S. oryzae*, and the red flour beetle, *T. castaneum* [89]. The Computerized Acoustical Larval Detection system was patented in 1988 and consists of a bowl capable of holding 1 L of grain sample with a diaphram at the bottom that is coupled with a microphone through a conduit [90]. The Ultrasonic Insect Detector, also known as the Purdue INsect Feeding monitor, was patented in 1990 [91]. The active sensor was able to detect the cowpea beetle, *Callosobruchus maculatus*, larva inside the cowpea seeds at the 40 kHz range [92] and monitor the development of the rice weevil, *Sitophilus o.* in maize kernels [93]. The sensor is only effective at distances less than a few millimeters because of the attenuation of the sound waves at high frequencies when traveling through grain [92].

The Sensor Technology and Applied Research (STAR) Center at Stevens Institute of Technology developed several iterations of the acoustic sensor systems before the current Acoustic Stored Product Insect Detection System. The first system featured a test container plated with ceramic piezoelectric sensors

3.1.4 Acoustic Insect Datasets

Several audio databases featuring insects have been published and are publically available. The Animal Sound Archive [94], InsectSet32 [95], InsectSet47 and InsectSet66 [85, 96], InsectSound1000 [97], and Singing Insects of North America (SINA) [98] all feature insect recordings; however, these datasets feature neither species commonly found infesting stored products nor audio recordings of insects within materials or stored product. BugBytes features 95 recordings of insects in different environments but only presents approximately 23 minutes of recordings with less than 15% of that pertaining to stored product insects; furthermore, only two of those recordings use piezoelectric disks [99].

3.2 Motivation

The motivation behind this research is to enhance the detection and monitoring of insects in stored products mainly to assist and speed up inspections performed by Custom and Border Patrol (CBP) at United States Ports of Entry. Although the sensors and associated algorithms were designed for this usage, the platforms can be adapted monitoring for stored product insects at other locations in the supply chain. The underlying technology and methods can also be expanded to service other materials. Beyond ensuring food safety and quality, effective monitoring can secure national agricultural, environmental, and ecological security from invasive species.

3.3 Methodology

3.3.1 Acoustic Stored Product Insect Detection System (A-SPIDS)

The Acoustic Stored Product Insect Detection System (A-SPIDS) was designed as a low-cost, portable, fast, and non-destructive alternative method to inspect stored products for insect infestations. The system is made up of a main body, an acoustically treated lid, and a base that provides vibration isolation and wireless charging to the integrated battery. The total system weighs 9 kg and has a length, height, and width of 30 cm, 24 cm, and 19 cm, respectively. The body of the system is 3D-printed ABS plastic lined with 1.875 cm thick acoustic foam and molded silicone. Figure **??** shows photographs of A-SPIDS in the open and closed configurations.

The main body features an integrated receptacle 26 cm long, 15 cm wide, and 3.5 cm tall, capable of holding 1 L to 1.5 L of product. The botton of the receptacle is lined with 32 piezoelectric discs arranged in four groups of eight. This container size not only satisfies inspection sampling requirements but also stays within

a b

Figure 1: The Acoustic Stored Product Insect Detection System in **??** open and **??** closed configurations

range for optimal insect detection based on the success of previous piezoelectric based sensors to detect the rice weevil, *Sitophilus oryzae*, and the adult red flour beetle, *Tribolium castaneum*, in grain up to 10 cm to 15 cm and 18.5 cm away, respectively, from the sensor [79, 100]. The maximum distance from the piezoelectric discs to any portion of the sample is set within the range even if the insects are active only on the surface of the substrate [101].

The piezoelectric discs are 27 mm in diameter, 0.33 mm thick, and made of piezoelectric ceramic lead zirconate titanate and brass [102]. The four groups correspond to channels zero through three in the recordings, complementary to their respective location in the system from left to right. The piezoelectric discs capture the mechanical sounds and vibrations generated by the insects moving, masticating, and mating within the product. Traditional acoustic microphones are placed throughout the body of the system to capture external noise and vibration. The microphones are POM-2735P-R with a diameter of 6 mm, a height of 4.7 mm, and a sensitivity of -35 ± 2 dB [103]. Microphone channels four and five are colocated with the piezoelectric channels zero and one, respectively, while channel six is placed within the system body, and channel seven is mounted on the exterior of the system. Figure **??** shows the layout of the sensors.

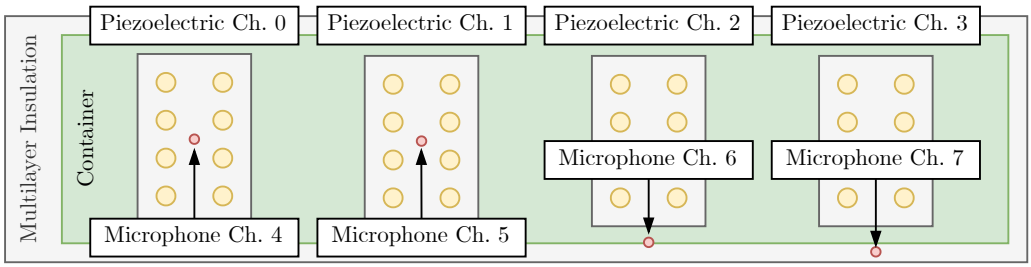

Figure 2: Diagram of A-SPIDS sensor layout

The main body also contains the supportive hardware for data acquisition, communication, and power. The piezoelectric discs and microphones output to a propriatary amplifier and then converted to an I2S stream using a Texas Instruments PCM1808 analog-to-digital converter (ADC) [104]. The data is collected by a

Raspberry Pi 3 A+ [105] and the I2S stream is digitized using a miniDSP MCHStreamer digital USB sound card [106] with a 24 kHz sampling rate. The Raspberry Pi transmits data to an aquisition computer through a Mikrotik mAP model RBmAP2nD [107]. The electronics are arranged on an electro-magnetic interference (EMI) shielding plate. Figure **??** diagrams the electronics in A-SPIDS.

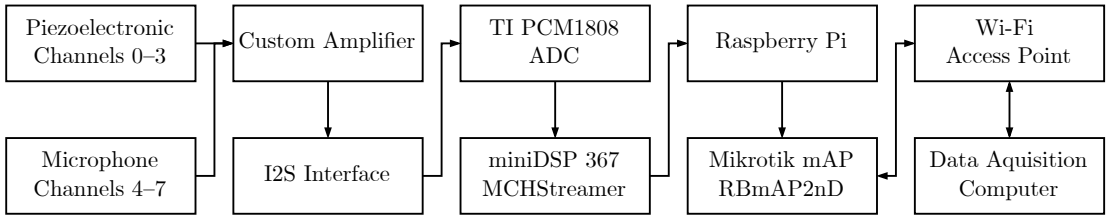

Figure 3: Diagram of A-SPIDS electronics

An integrated battery can maintain the system for several hours and recharges using a reciever coil when positioned on the platform base which is powered by a 24 V power supply. The sensor can be lifted off and operated without the charging platform.

3.3.2 Tested Insect, Materials, and Noise Levels

Although stored products insects proliferate across a wide range of commodities, they are difficult to obtain for research purposes. The cowpea beetle, *Callosobruchus maculatus*, the confused flour beetle, *Tribolium confusum*, and the mealworm beetle, *Tenebrio molitor*, are commercially available and were chosen based on their similarities to the sizes, qualities, and behavior to the species targeted during inspection.

The cowpea weevil or cowpea seed beetle and also known as the pulse beetle, *Callosobruchus maculatus*, is part of the leaf beetle family, *Chrysomelidae*, and infests cowpea and mung beans. Adults are small, round beetles, browninish with black, pale spots and stripes, and range from 2 mm to 4 mm in length. Adults have a lifespan of 12 days and colonies reproduce until their food supply is depleted. Females lay eggs on the surface of the beans. Upon hatching in around five days, larvae dig into the seed and feed on the cotyledons as they develop until they bore out of the bean as adults within 25-30 days [108]. This boreing behavior is similar to other internal feeders, such as the lesser grain borer, *Rhyzopertha dominica*, and the rice weevil, *Sitophilus oryzae*, that are targetted as problematic by the USDA. Flying versions of the species can disperse to new locations, including directly to cowpea fields. Even a 1 % to 2 % infestation on a field can result in 80 % loss of product after six to eight months of storage. In West Africa, up to 100 % loss was reported after several months of unmitigated infestation [48].

The confused flour beetle, *Tribolium confusum*, is an arthropod that infests farinaceous products such as rice and flour. They are reddish-borwn with clubbed anthennas and range from 2 mm to 4 mm in length. As a secondary pest, the larvae and adult forms are found in damaged and processed grains. Their infestations often result in a grey tint, odor, production of toxic quinones, and encourage mold growth. The confused flour beetle eggs are white and microscopic, proving a challenge to detect among the material they settle. Their lifecylce takes 40 to 90 days and the adults are capable of living up to three years in moderate climate conditions [48, 109].

The *Tenebrio molitor* is known in its larval form as the mealworm and in its adult form as the darkling beetle. The larvae are 1 cm to 2 cm long, have a dark brown head, and a segmented yellow body with a hard exoskeleton. Adults also grow up to 2 cm in length but are a shiny brown or black color. Females can lay up to 6000 eggs during their lifecycle. Darkling beetles release benzoquinones that result in an unpleasant odor [48, 110]. Table **??** provides a summary of the tested insect species.

Table 1: Description of tested live specimens

Scientific Name	**Common Name**	**Size** [mm]	**Weight** [mg]	**Density** [$\mathrm{kg\,L^{-1}}$]
C. maculatus	Cowpea beetle	$4 \times 1 \times 1$	2.1	0.525
T. confusum	Confused flour beetle	$4 \times 2 \times 2$	1.0	0.625
T. molitor	Darkling beetle	$15 \times 5 \times 3$	160.0	0.711
T. molitor larva	Mealworm	$20 \times 2 \times 2$	130.0	1.625

A variety of stored product materials were tested with and without insect specimen including oatmeal, *Avena sativa*, rice, *Oryza sativa*, wheat in the form of whole groat kernels and processed flour, *Triticum aestivum*, and corn in the form of Corn Flakes, *Zea mays*. The material samples were kept in sealed storage containers to prevent contamination and infestation. During testing, 1 L of clean material was transferred into a plastic bag and placed into the sensor container. After several minutes of silence were recorded to ensure the grains were settled, a single insect was introduced into the material. Table **??** summarizes the duration of recordings for each insect-material pairing. The insect-material pairings are not fully balanced as a result of varying life stages and availability of the insects, the need to test in clean and uncontaminated materials, and the time and budget restraints of the project.

Table 2: Duration of recordings in minutes for insect-material pairings with silence. The duration of recordings with external noise are accommodated with parenthesis.

	Avena sativa	***Oryza sativa***	***Triticum aestivum***		***Zea mays***
	Oatmeal	White Rice	Wheat Groats	Flour	Corn Flakes
C. maculatus	12	258 (12 noise)	13	20 (1 noise)	17
T. confusum	39	274 (16 noise)	189	55 (1 noise)	16
T. molitor	68	15	9	8	13
T. molitor larva	46 (4 noise)	268 (132 noise)	51 (2 noise)	128	43 (5 noise)
No insect	4	219 (64 noise)	142 (2 noise)	24	23

The expected usage of the system would be within a variety of environments such storage facilities, warehouses, and ports, where machinery, vehicles, and people can create significant background noise. The sound pressure levels (SPL) can range from 35 dBA to 50 dBA common for an office up to 110 dBA for a factory [68, 111]. The system was exposed to varying levels of natural and simulated, ambient and articial noise with and without live insect specimens to examine the efficacy of the sound insulation as well as the influence on the detection, false positive, and false negative performance. The ambient noise was a result of the surrounding laboratory environment including equipment, HVAC units, and personnel as well as vehicular and aircraft traffic. Artificial noise, such as pre-recorded sweeps, white noise, loud conversations, crowds, and factory noise were played through a speaker system at increased steps from 60 dBA to 110 dBA and the noise levels were measured using a calibrated sound level meter, the Extech 407730 Digital Sound Level Meter manufactured by Teledyne FLIR [112–115].

3.3.3 Stored Product Insect Database (SPIDB)

The Stored Product Insect Database (SPIDB) provides interface to access the data collected from the A-SPIDS platform. The database extends the Sound Organization for Network Integration and Collection/Collaboration (SONICDB) framework, by adding the `Material` model and additional attributes to the `Event` and `Subject` models. Figure **??** shows the schema for the SPIDB:

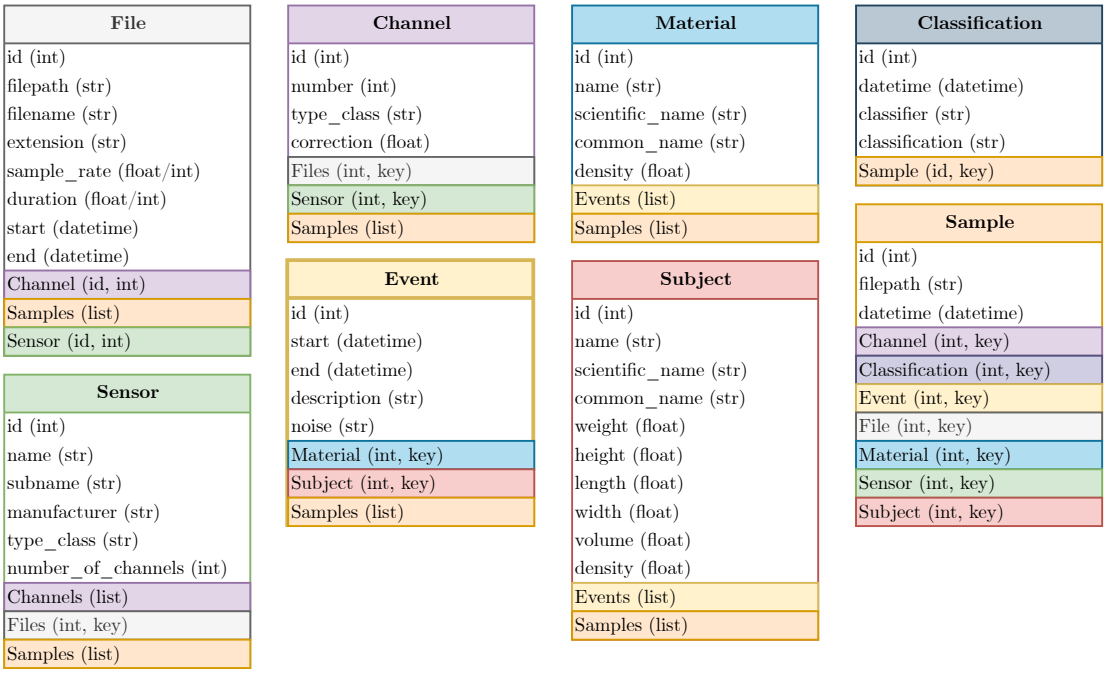

Figure 4: Schema of the Stored Product Insect Database (SPIDB)

The Stored Product Insect Database (SPIDB) is available on GitHub with supporting documentation [116]. The associated dataset is available on Kaggle [117].

4 Back Matter

4.1 Research Timeline

Completed Tasks

- **February-May 2022** Literature review and analysis of acoustic detection data from 2021 and 2022 field tests. Preparation for conference presentation and publication.
- **March 2022** Ph.D. Qualification Exam
- **May 2022** ASA Conference, Denver, CO, USA - Improvements to the Stevens Drone Acoustic Detection System
- **February-December 2023** Development of the Acoustic Stored Product Insect Detection System (A-SPIDS) and collection of vibro-acoustic signatures of various insects in stored products.
- **January 2024** Literature review and analysis of vibro-acoustic signatures of various insects in stored products. Preparation for publication.
- **March 2024** AIMS Conference, New York, NY, USA - Perspectives of Machine Learning for Acoustic Methods of Stored Product Insect Detection
- **January 2025** Publication of *Vibro-Acoustic Signatures of Various Insects in Stored Products* in MDPI Sensors
- **January-August 2025** Development and validation of stored product insect detection algorithm
- **August 2025** DHS RDT&E Conference, Washington, D.C., USA - Stored Product Insect Detection System (SPIDS)

Current Tasks

- **August 2025** Dissertation Proposal Defense
- **September 2025** Finish writing *Acoustic Detection of Stored-Product Insects Under High Ambient Noise Conditions* to MDPI Sensors
- **October 2025** Finish writing *The Drone Acoustic Database: A Rich Multi-Sensor Dataset for Acoustic UAV Detection, Tracking, and Classification* to MDPI Data
- **November 2025** Work on dissertation

Planned Tasks

- **September 2025** Submit *Acoustic Detection of Stored-Product Insects Under High Ambient Noise Conditions* to MDPI Sensors
- **October 2025** Submit *The Drone Acoustic Database: A Rich Multi-Sensor Dataset for Acoustic UAV Detection and Classification* to MDPI Data
- **November 2025** Work on dissertation
- **End of December 2025** Dissertation Defense

4.2 Publications and Conferences

Publications

- **Vibro-Acoustic Signatures of Various Insects in Stored Products**

  D. Kadyrov, A. Sutin, N. Sedunov, A. Sedunov, and H. Salloum, "Vibro-Acoustic Signatures of Various Insects in Stored Products," Sensors, vol. 24, no. 20, Art. no. 20, Jan. 2024, doi: 10.3390/s24206736.

Planned Publications

- Acoustic Detection of Stored-Product Insects Under High Ambient Noise Conditions
- The Drone Acoustic Database: A Rich Multi-Sensor Dataset for Acoustic UAV Detection, Tracking, and Classification

Conferences

- **Improvements to the Stevens Drone Acoustic Detection System**

  Acoustic Society of America (ASA) 2022, Denver, CO, USA

  D. Kadyrov, A. Sedunov, N. Sedunov, A. Sutin, H. Salloum, and S. Tsyuryupa, "Improvements to the Stevens Drone Acoustic Detection System," Proceedings of Meetings on Acoustics, vol. 46, no. 1, p. 045001, Aug. 2022, doi: 10.1121/2.0001602.

- **Perspectives of Machine Learning for Acoustic Methods of Stored Product Insect Detection**

  Artificial Intelligence Meets Science (AIMS) Conference, 2024, New York, NY, USA

- **Stored Product Insect Detection System (SPIDS)**

  Department of Homeland Security (DHS) Research and Development, Testing and Evaluation (RDT&E) Conference, 2025, Washington, D.C., USA

4.3 Policy and Regulatory Considerations

The research and products developed as part of the unmanned aerial vehicle study were funded by the Air Force Research Laboratory (AFRL) under contract AFRL-2022-3230. Prior to public release, all publications and work products need to be reviewed and approved by AFRL Public Affairs Office. The drone flights were conducted in accordance with the Federal Aviation Administration (FAA) regulations under Part 107.

The research and products developed as part of the stored product insect detection study were funded by the Cross-Border Threat Screening and Supply Chain Defense (CBTS) under grant 18STCBT00001. This office no longer exists and the publications and work no longer need to be reviewed and approved by this office. The insects used in the study are non-invasive and non-harmful to the environment, and the research was conducted in accordance with the Institutional Animal Care and Use Committee (IACUC) guidelines. The insects were disposed of after the completion of the study following the IACUC and USDA guidelines.

Daniel Kadyrov is currently a full-time employee of the National Urban Security Technology Laboratory (NUSTL), within the Science and Technology Directorate (S&T), a part of the Department of Homeland Security (DHS). Publications and work products need to be reviewed and approved by the DHS S&T ethics office and the NUSTL Director prior to public release.

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